House Price Prediction Using Linear Regression

Artificial Intelligence & Machine Learning – Task 1

1. Introduction

This project introduces the complete machine learning workflow for a house price prediction problem. The workflow includes data loading, exploration, preprocessing, model training, evaluation, and reporting. A Linear Regression model is trained using the California Housing Dataset to predict median house values.

2. Objective

The objective is to build and evaluate a Linear Regression model for predicting house prices using the California Housing Dataset. The project also demonstrates exploratory data analysis, visualization, train-test splitting, model training, prediction, and evaluation using MAE, RMSE, and R^2.

3. Dataset Description

The California Housing Dataset is used for this project. It contains numerical information describing housing areas in California. The target variable is MedHouseVal, representing the median house value. The dataset can be accessed directly through scikit-learn.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the structure and characteristics of the dataset. The analysis includes checking the dataset shape, column names, data types, descriptive statistics, and missing values. Visualizations such as histograms, scatter plots, and a correlation heatmap help identify distributions and relationships among variables.

5. Data Preparation

The dataset is divided into independent variables (features) and the dependent variable (target). MedHouseVal is selected as the target, while the remaining columns are used as input features. The data is then divided into training and testing sets using an 80:20 split. The training set is used to train the model, while the test set is used to evaluate its performance on unseen data.

6. Linear Regression Model

Linear Regression is a supervised machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables. In this project, the Linear Regression model learns relationships between the housing features and the median house value from the training data.

7. Model Training and Prediction

After splitting the dataset, a Linear Regression model is created and trained using the training data. The trained model is then used to generate predictions for the test dataset. Actual and predicted values can be compared to understand the prediction performance.

8. Model Evaluation

The model is evaluated using three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. MAE represents the average absolute difference between actual and predicted values. RMSE gives greater weight to larger errors. R^2 indicates how well the model explains the variation in the target variable.

Evaluation Metric	Purpose
MAE	Measures the average absolute prediction error.
RMSE	Measures prediction error while giving more weight to larger errors.
R^2 Score	Indicates how well the model explains variation in the target variable.

9. Results

The Linear Regression model is successfully trained and evaluated on the California Housing Dataset. The notebook calculates MAE, RMSE, and R^2 using the test data. The exact metric values should be taken from the executed notebook so that the report reflects the user's actual model results.

10. Improvement Ideas

The model can potentially be improved by performing additional feature engineering, trying alternative regression algorithms such as Ridge or Lasso Regression, using Random Forest or Gradient Boosting methods, and performing hyperparameter tuning. These approaches can be compared with the Linear Regression baseline.

11. Conclusion

A complete machine learning workflow was implemented for house price prediction using the California Housing Dataset. The project covered data exploration, visualization, train-test splitting, Linear Regression model training, prediction, and evaluation. The use of MAE, RMSE, and R^2 provides a structured way to assess the model's performance and identify possible improvements.